

# Mathematics for Engineers I - MAT 1166

## Exam Preparation Guide

### General Preparation Tips

- **Understand Exam Format:**

- Since this is a 10-credit module, the total duration will be 2 hours.

- **Approved Materials:**

As this is your first exam in your first semester at the University of Rwanda, you need to keep the following in mind.

- No written materials or electronic devices are permitted in the exam room. This means that you cannot use your (smart) phones as calculators!
- Only (non-programmable) calculators are allowed during the exam.

### Focus Areas for preparation

For better preparation for your final exam, please focus on reading your syllabus, solving all exercises in the syllabus, reworking questions from assignments, quizzes, and CATs, and also conducting research on additional exercises and applications based on the following key topics

- ✎ **Calculations on arithmetic and geometric Progression**
- ✎ **Calculations on limit and Continuity of functions**
- ✎ **Derivatives and their applications**
- ✎ **Integrals, technics of integrations, and applications of integrals**
- ✎ **Improper Integrals**
- ✎ **Operations on matrices**
- ✎ **Vectors in 3D Space, dot and cross products and applications**
- ✎ **Series and Expansions, tests of convergence of series, find radius and intervals of convergence of power series and Taylor/Maclaurin polynomials**
- ✎ **Systems of Linear Equations, resolution methods and real-world applications**
- ✎ **Eigenvalues and Eigenvectors of a matrix computation, and relate them to real-world applications**